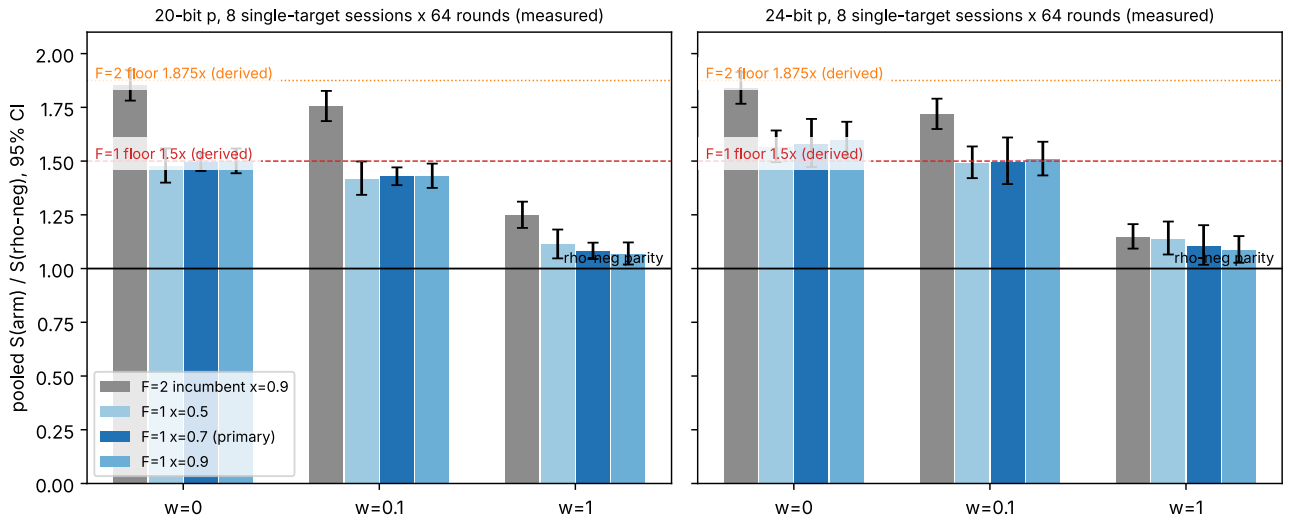
One-point (F = 1) walk-table oracle vs the F = 2 incumbent,

single-target ecbench (2026-11-07) **Class.** Measured, single-target ecbench sessions under a protocol frozen before data (PROTOCOL.md hashes in `data/frozen.sha256`). Counted units only (ecbench GAE, S = cost/sqrt(r)), every comparison at w = 0, 0.1, 1. Public synthetic Curve25519-family miniatures; no security claim. No pcc claim has been filed for these numbers. Each one is an unverified assertion until it is reviewed [pcc:unverified].

Question. The lab's generic incumbent `wwalk` (`x=0.9,g=1,grow=2,seed=run` on a 2-abscissa base) has F = 2 unknown base logarithms. Does a one-seed walk table (F = 1) cost less, end to end, for a single target? [pcc:unverified]

Answer. Yes. At k = 20 the primary F = 1 arm (`f1-x07`) costs 0.807x the F = 2 incumbent at w = 0, with 95 % interval [0.777, 0.839]. Its interval stays below 1 at w = 0.1 and at w = 1 [pcc:unverified]. At k = 24 the F = 1 arm costs 0.859x [0.819, 0.901] at w = 0 [pcc:unverified]. Rho-neg remains ahead. At k = 20 the remaining distance is 1.50x at w = 0, 1.43x at w = 0.1 and 1.08x at w = 1, and every interval excludes parity [pcc:unverified]. The w = 0 distance equals the derived F = 1 collision floor of 1.5x [pcc:unverified].



1. What changed in the code

`WalkTable::build_with` (`src/cryptanalysis/ic_boundary.rs`) now accepts a factor base of exactly one abscissa when `g=1`. In that mode the jumps are `[a_j]P1 + [c_j]G` and the entries are `u*P1 + z*G`. A verified key collision gives the one-column affine row `du*s1 = -dz`, which pins `log P1`. A target probe hit gives the target's row. `grow=1|2` keeps walking until the verified rows reach rank 1, capped at 2T or 3T. A one-seed base without `g=1` is refused, because its row would be empty. The 2-seed path keeps its random draws, walk and rows unchanged. The base is specified as `prime-projected-abscissa:size=1,scan_cap=32`. The ecbench help text and the oracle documentation say so.

Tests (`data/cargo-test-wwalk.txt`: 7 passed, 0 failed; `data/cargo-test-plan-refuses.txt`: 6 passed) include the existing wwalk tests, unchanged, and the new `wwalk_one_seed_rows_are_affine_and_grow_reaches_rank_one`. That test checks the charging ledgers, `v = 0`, one-column rows that hold under brute-force logs, probe answers that rebuild the target, growth stopping at rank 1, and the refusals.

2. Design (as frozen)

- Binary: release build of this tree in `/home/user/ctarget-f1` (`data/binary.sha256`). The existing `/home/user/ctarget` binary is untouched.
- 8 sessions per curve size, each with `targets_per_curve = 1` (one public target per session) and its own target and measurement seeds (`scripts/make_specs.py`, `specs/`). Each arm ran 64 measured rounds plus 1 warm-up, in alternate order.
- Arms: `rho-neg` (reference), `rho-aa` (A/A), and `f2-inc` (F = 2 incumbent: size=2 base, `wwalk:x=0.9,g=1,grow=2,seed=run`). The candidates `f1-x05`, `f1-x07` (primary) and `f1-x09` use a size=1 base and `wwalk:x=X,g=1,grow=2,seed=run`. All pipeline arms use `targets=stride-generator`.
- Audit: `ecbench verify --replay 1` for every session (`data/verify-k*-s*.txt`).
- Pooling: `scripts/pooled_compare.py`, copied unchanged from the T12 study. It takes the ratio of per-session sums with a linearised t interval and df = 7 (one session = one target = one cluster). `scripts/analyse.py` writes `data/pooled.json`.
- Runtime: about 52 s of sessions at k = 20 and about 60 s at k = 24. Both sizes ran in full, well inside the 90-minute cap.

3. Results (measured)

Every session completed. Each wrote 390 of 390 records, all verified, and its audit reproduced 1 of 1 replays with 0 problems (`data/verify-*.txt`) [pcc:unverified].

Mean S per arm (3,120 records per size; source `data/pooled.json`, which pools `sessions/k*-s*/records.jsonl`):

k	arm	S w=0	S w=0.1	S w=1
20	rho-neg	1.552	1.664	2.672
20	f2-inc (F=2)	2.875	2.921	3.337
20	f1-x05	2.293	2.361	2.973
20	**f1-x07**	2.321	2.378	2.891
20	f1-x09	2.328	2.381	2.856
24	rho-neg	1.161	1.265	2.205
24	f2-inc (F=2)	2.134	2.174	2.532
24	f1-x05	1.819	1.888	2.513
24	**f1-x07**	1.834	1.894	2.438
24	f1-x09	1.856	1.910	2.398

The F = 2 incumbent at k = 24 (2.134) agrees with the rebaseline-seedrun figure of 2.105 for the same arm on the same curve [pcc:unverified].

Pooled ratios with 95 % intervals over 8 single-target sessions (`data/pooled.json`):

k	ratio	w = 0	w = 0.1	w = 1
20	f1-x07 / f2-inc	0.807 [0.777, 0.839]	0.814 [0.782, 0.847]	0.866 [0.820, 0.915]
20	f1-x05 / f2-inc	0.798 [0.775, 0.821]	0.808 [0.786, 0.831]	0.891 [0.865, 0.918]
20	f1-x09 / f2-inc	0.810 [0.774, 0.848]	0.815 [0.777, 0.855]	0.856 [0.801, 0.915]
20	f1-x07 / rho-neg	1.495 [1.454, 1.538]	1.429 [1.388, 1.471]	1.082 [1.044, 1.120]
20	f2-inc / rho-neg	1.852 [1.781, 1.925]	1.755 [1.686, 1.826]	1.249 [1.189, 1.311]
20	rho-aa / rho-neg	1.000 [1.000, 1.000]	1.000	1.000
24	f1-x07 / f2-inc	0.859 [0.819, 0.901]	0.871 [0.829, 0.916]	0.963 [0.899, 1.032]
24	f1-x05 / f2-inc	0.852 [0.820, 0.886]	0.869 [0.833, 0.906]	0.992 [0.934, 1.054]
24	f1-x09 / f2-inc	0.870 [0.844, 0.896]	0.879 [0.851, 0.907]	0.947 [0.903, 0.993]
24	f1-x07 / rho-neg	1.580 [1.472, 1.696]	1.497 [1.393, 1.610]	1.106 [1.018, 1.202]
24	f2-inc / rho-neg	1.839 [1.767, 1.914]	1.718 [1.649, 1.790]	1.148 [1.093, 1.207]

4. Predictions (PROTOCOL.md section 5)

id	prediction	outcome	status
P1	k=20 f1-x07/f2-inc upper end < 1 at w = 0, 0.1, 1	0.839, 0.847, 0.915	**held** [pcc:unverified]
P1b	k=20 f1-x07/f2-inc point in [0.80, 0.95] at w = 0	0.807	**held** (near the lower edge) [pcc:unverified]
P2	k=20 f1-x07/rho-neg point in [1.45, 1.85] at w = 0	1.495	**held** [pcc:unverified]
P3	k=20 f1-x07/rho-neg lower end > 1 at all w (no parity)	1.454, 1.388, 1.044	**held** [pcc:unverified]
P4	rho-aa/rho-neg interval contains 1	1.000 exactly	**held, but degenerate** [pcc:unverified]
P5	k=20 f1-x05 and f1-x09 / f2-inc upper end < 1 at w = 0	0.821, 0.848	**held** [pcc:unverified]
P6	k=24 f1-x07/f2-inc upper end < 1 at w = 0	0.901	**held** [pcc:unverified]

P4 is degenerate. The rho-aa and rho-neg records are identical, run for run, because ecbench derives the run seed from the workload and round, not from the arm name. The A/A arm therefore gives no noise floor. The real noise is the between-session spread, and the t intervals carry it [pcc:unverified]. Not scored, but reported: at k = 24 and w = 1, two of the three F =

1 / F = 2 intervals include 1.00, so the k = 24 ordering holds at w = 0 and w = 0.1 but is unresolved at w = 1 for x = 0.5 and 0.7 [pcc:unverified].

5. Distance to matched rho (rho-neg, same sessions)

k	w = 0	w = 0.1	w = 1
20, F = 1 (f1-x07)	1.50x	1.43x	1.08x
20, F = 2 (f2-inc)	1.85x	1.76x	1.25x
24, F = 1 (f1-x07)	1.58x	1.50x	1.11x
24, F = 2 (f2-inc)	1.84x	1.72x	1.15x

The F = 1 arm closes about 40 % of the F = 2 incumbent's w = 0 gap to rho at k = 20 ((1.852 - 1.495) / 0.852, derived) and about 31 % at k = 24 ((1.839 - 1.580) / 0.839, derived) [pcc:unverified]. The gap narrows at w = 1 because rho-neg's unpriced native work, about one uncharged negation canonicalisation per walk step (`canonicalisations_uncharged`), is larger per GAE than the walk-table arms' lookups and row operations [pcc:unverified].

6. Reading and limits

- **What the gain is.** F = 1 needs two collisions: one walk self-collision to pin `log P1`, then one table hit for the target. F = 2 needs three. The measured gain is the drop from three collisions to two. The frozen model (PROTOCOL.md section 4) predicted a ratio of 0.876 (derived). The measured ratio is 0.807 at k = 20 and 0.859 at k = 24 [pcc:unverified].
- **How close this is to rho.** F = 1 is the last index-calculus step before F = 0. F = 0 (a base of known logarithms) is rho or BSGS relabelled and is excluded by the success rule. The F = 1 arm still has a base element of unknown logarithm, solved by its own collision. It is not rho relabelled, but the factor base does no decomposition work: the arm is a two-collision generic algorithm. Whether the lab counts it as an index-calculus improvement or as a step toward relabelling is a call for the reviewer. This README makes no leaderboard claim [pcc:unverified].
- **The floor still binds.** The w = 0 distance of 1.495x sits on the derived F = 1 collision floor of 1.5x. Under the generic model, no generic F = 1 variant can go much lower. The k = 20 rho-neg S of 1.55 is far above the model's 0.886, because small-r overheads inflate both arms. One sampled k = 20 rho-neg record has 54 set-up additions and 76 set-up doublings beside 123 walk operations. At k = 20 the ratio therefore matches the floor partly by cancellation of these overheads [pcc:unverified].
- **Model gap.** The asymmetric walk-table model predicted 1.75x rho-model for F = 1 and 1.99x for F = 2 (derived). The measured ratios to measured rho-neg are lower because rho-neg's measured overhead factor is larger than the walk-table arms' [pcc:unverified].
- **Not done.** The study has no VALIDATION.json, no replay certificate or validator report, and no leaderboard or `benchmarks/README.md` row. Those are for the landing session under the validation gate. The raw `records.jsonl` files are in `data/sessions-k{20,24}-records.tar.gz`, which is git-ignored per session, and their hashes are in the benchrec records.

Files

`PROTOCOL.md`, `specs/`, `scripts/{make_specs.py,run_sessions.sh,pooled_compare.py,analyse.py,make_figure.py}`, `sessions/k*-s*/{session,spec,plan,host}.json`, `data/` (plans, run logs, bands, verify reports, `pooled.json`, `binary.sha256`, `frozen.sha256`, `cargo-test-wwalk.txt`, record tarballs), `figures/ratio_to_rho.{svg,png}`, `REPORT.pdf`. Benchmark records: `benchmarks/records/20261107T1945Z-curve25519-f1-walk-k20.json` and `benchmarks/records/20261107T1947Z-curve25519-f1-walk-k24.json`.